

Experiment No. 4

Experiment Name

Frequency Domain Image Filtering using Fourier Transform

Aim

To implement image filtering in the frequency domain using the Fourier Transform and analyze the effectiveness of frequency-based filtering techniques for noise removal, image enhancement, and feature preservation using Python and OpenCV.

Purpose

Many image processing operations can be performed more efficiently in the frequency domain than in the spatial domain. The Fourier Transform converts an image from the spatial domain into its frequency components, enabling selective manipulation of low-frequency and high-frequency information. Frequency domain filtering is widely used in applications such as medical image enhancement, satellite image analysis, image restoration, and biometric systems.

In this experiment, students will apply the Discrete Fourier Transform (DFT) to an image, analyze its frequency spectrum, and implement Low-Pass and High-Pass Frequency Filters to observe their effects on image quality and feature extraction.

Prerequisites

- Basic knowledge of Digital Image Processing
- Understanding of spatial and frequency domains
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), NumPy, and Matplotlib
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Import the required libraries and load a grayscale image for processing.
2. Compute the Discrete Fourier Transform (DFT) of the image using OpenCV or NumPy.
3. Shift the zero-frequency component to the center of the frequency spectrum for visualization.
4. Display and analyze the magnitude spectrum of the transformed image.

5. Design and apply a Low-Pass Frequency Filter to suppress high-frequency components and reduce image noise.
6. Design and apply a High-Pass Frequency Filter to suppress low-frequency components and enhance edges and fine details.
7. Perform the Inverse Fourier Transform (IDFT) to reconstruct the filtered images in the spatial domain.
8. Compare the original image, frequency spectrum, low-pass filtered image, and high-pass filtered image.
9. Analyze the impact of frequency domain filtering on image smoothness, edge enhancement, and noise removal.
10. Document the observations and discuss the advantages of frequency domain filtering in practical computer vision applications.

Questions

1. What is the Fourier Transform? Why is it important in digital image processing?
2. Differentiate between the spatial domain and the frequency domain.
3. What is the significance of the Discrete Fourier Transform (DFT) in image processing?
4. Explain the purpose of shifting the zero-frequency component to the center of the frequency spectrum.
5. Compare Low-Pass Frequency Filters and High-Pass Frequency Filters with suitable applications.
6. What is the role of the Inverse Fourier Transform (IDFT) in image reconstruction?
7. Why is frequency domain filtering preferred for certain image enhancement tasks?
8. Mention four real-world applications where Fourier Transform is used in computer vision and image analysis.
9. Compare frequency domain filtering with spatial domain filtering based on computational efficiency and practical applications.
10. How does frequency domain filtering improve the performance of image restoration and feature extraction techniques?